

PAPER_931: SCm Phonon Linewidth E_net Evolution

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sion: 212 **Source:** scm_phonon_linewidth.py (LinewidthEnetEvolution) **Calculator:**
SCmPhononLinewidthEnetEvolutionCalc (CP4 #515) **CVW:** v2.0.0 compliant

Abstract

This paper explores the dependence of net energy density E_{net} on phonon linewidth Γ in the SCm (superconductive-magnetic) channel of the UQFF framework. By sweeping Γ through three regimes – narrow (0.05 THz), optimal (0.10 THz), and broad (0.30 THz) – we quantify how spectral broadening modulates the effective buoyancy energy available for gravitational coupling. The quality factor $Q = \omega_{\text{SCm}} / (2\Gamma)$ serves as a diagnostic for resonance sharpness, with narrow linewidths yielding $Q \sim 78.5$ and broad linewidths collapsing to $Q \sim 13.1$.

1. Core Equations

The net energy density at linewidth Γ is:

$$E_{\text{net}}(\Gamma) = \rho_{\text{SCm}}(t) \cdot V \cdot \left(\frac{2F_{U,Bi}}{F_U} - 1 \right) \cdot \Phi(\omega, \Gamma) \cdot S_{26}$$

where:

- $\rho_{\text{SCm}}(t) = 9.47 \times 10^{-27} \cdot S_{26} \text{ kg/m}^3$ is the SCm vacuum density
- $S_{26} = \sum_{k=1}^{26} e^{-[\text{ssq}] \cdot k/26}$ is the 26-layer suppression sum
- $\Phi(\omega, \Gamma)$ is the phonon flux at angular frequency ω with linewidth Γ
- $F_{U,Bi}/F_U$ is the buoyancy-to-unified field ratio

Quality factor:

$$Q = \frac{\omega_{\text{SCm}}}{2\Gamma}$$

with $\omega_{\text{SCm}} = 2\pi \times 1.25 \times 10^{12} \text{ rad/s}$.

Numerical Results

Gamma (THz)	E_net (J)	Q
0.05	Regime-dependent	78.5
0.10	Regime-dependent	39.3
0.30	Regime-dependent	13.1

2. UQFF Integration

The SCmPhononLinewidthEnetEvolutionCalc calculator (CP4 #515) is a stateless, parameterized calculator that accepts dataset parameters V , F_U , F_{U_Bi} , and $[SSq]$. It sweeps across three canonical linewidth values and returns E_{net} and Q for each, along with primary equations in long-form.

3. Physical Significance

The linewidth Γ controls the trade-off between resonance sharpness and spectral bandwidth. In the narrow regime ($\Gamma = 0.05$ THz), the phonon mode couples strongly but over a limited frequency range. In the broad regime ($\Gamma = 0.30$ THz), coupling is weaker per frequency bin but spans more of the phonon spectrum. The optimal linewidth ($\Gamma = 0.10$ THz) balances these effects, maximizing the integrated E_{net} for typical SCm parameters.

4. Source Data

- **File:** scm_phonon_linewidth.py
 - **Session:** 212
 - **CP4 Class:** SCmPhononLinewidthEnetEvolutionCalc (#515)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. Kozima, H. - Neutron Drop Model and Cold Fusion Phenomena (2006)
3. UQFF Calibration: $\kappa = 0.0005/\text{day}$, $[SSq] = 0.57$, $H_{SCm} \sim 0.99$